

(51) International Patent Classification:

A47J 31/18 (2006.01) A47J 31/00 (2006.01)

(21) International Application Number:

PCT/EP2019/054639

(22) International Filing Date:

26 February 2019 (26.02.2019)

(25) Filing Language:

English

(26) Publication Language:

English

(30) Priority Data:

A 2018/04613 02 April 2018 (02.04.2018) TR

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(81) Designated States (unless otherwise indicated, for every  
kind of national protection available): AE, AG, AL, AM,  
AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ,  
CA, CH, CL, CN, CO, CR, CU, CZ, DE, DJ, DK, DM, DO,  
DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN,  
HR, HU, ID, IL, IN, IR, IS, JO, JP, KE, KG, KH, KN, KP,  
KR, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, ME,  
MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ,  
OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA,  
SC, SD, SE, SG, SK, SL, SM, ST, SV, SY, TH, TJ, TM, TN,  
TR, TT, TZ, UA, UG, US, UZ, VC, VN, ZA, ZM, ZW.

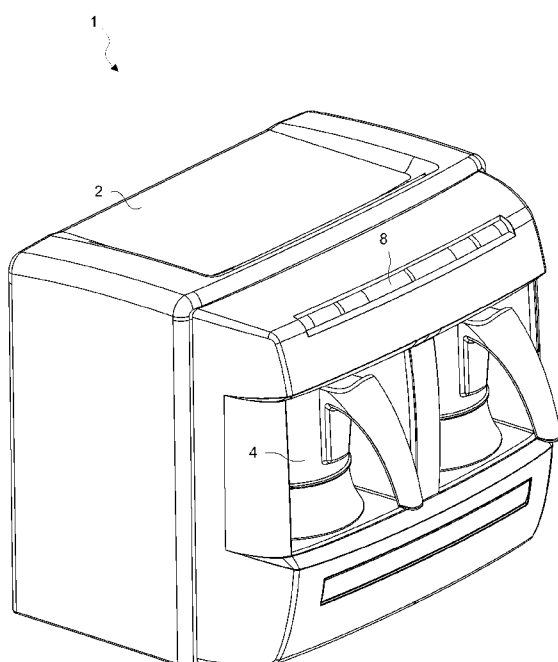
(84) Designated States (unless otherwise indicated, for every  
kind of regional protection available): ARIPO (BW, GH,  
GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ,  
UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, RU, TJ,  
TM), European (AL, AT, BE, BG, CH, CY, CZ, DE, DK,  
EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV,  
MC, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM,  
TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW,  
KM, ML, MR, NE, SN, TD, TG).

Published:

— with international search report (Art. 21(3))

(54) Title: A COFFEE MACHINE

Figure 1



(57) Abstract: The present invention relates to a coffee machine (1) comprising a body (2), a cooking compartment (3) provided in the body (2), a cooking chamber (4) in which the cooking process is performed, placed in the cooking compartment (3), a heater (5) provided on the base of the cooking compartment (3), and a control unit (6) starting the cooking process by activating the heater (5), terminating the cooking process by turning the heater (5) off, and enabling controlling the cooking process.

**Description****A COFFEE MACHINE**

- [0001] The present invention relates to a coffee machine capable cooking Turkish coffee.
- [0002] Coffee machines capable of automatically cooking Turkish coffee are known in the art. In such coffee machines, various sensors such as heat, infrared or conductivity sensors are used in terminating the cooking process. In applications where infrared sensors are used, it is almost possible to cook coffee with identical taste and consistency every time, but the sensor surface becomes dirty after several cooking processes, requiring manual cleaning by the user. Otherwise the infrared sensor may make incorrect determinations. In addition, infrared sensors do not measure distances directly, and they determine a given distance according to the intensity of the retroreflected light. Distance determination requires each sensor to be calibrated separately, increasing labor costs in production stage. The incapacity of the infrared sensors to directly measure distances also leads to problems in cooking different types of Turkish coffee; in lighter and darker colored coffees, an infrared sensor reads different distances even when the actual foam height is equal because of the variation in the reflected light intensity, thereby making it not possible to cook coffee with identical taste every time. Furthermore, the optimum operational range of infrared sensors is narrow, thereby restricting the depth of cooking chambers available in Turkish coffee machines using infrared sensors and therefore the amount of coffee which can be cooked simultaneously. Although heat or conductivity sensors are less affected from dirt, it is not possible to cook coffee with identical taste and consistency every time since such sensors have low precision. The consumers demand a coffee machine in which cooking is sensed by a sensor capable of making high-precision measurements and not requiring separate calibrations for different types of coffee.
- [0003] State of the art European patent application no. EP1763311 discloses a Turkish coffee machine in which whether coffee is cooked is sensed by using an infrared sensor.

- [0004] State of the art European patent application no. EP2316312 discloses a coffee machine in which whether Turkish coffee is cooked is sensed by using a conductivity sensor.
- [0005] The aim of the present invention is to realize a coffee machine not requiring different calibrations for different types of coffee and capable of cooking a desired amount of Turkish coffee with identical taste and consistency every time.
- [0006] The coffee machine realized to achieve the aim of the present invention and disclosed in the first claim and the dependent claims, comprises a time of flight sensor provided in a cooking compartment in which a cooking chamber is placed where coffee cooking is performed, and a control unit deciding to start and/or to terminate the cooking process according to data received from the time of flight sensor. Since a time of flight sensor enables measuring distances directly, it allows minimizing the failure rate in cooking coffees of varying color tones. In addition, a cooking chamber of a desired size (depth) can be used thanks to a time of flight sensor having a wider operational range, enabling cooking a desired number of cups of coffee simultaneously.
- [0007] In an embodiment of the invention, the time of flight sensor begins measuring upon a user pressing a cooking start button. In another embodiment of the invention, the time of flight sensor begins measuring upon placement of the cooking chamber in the cooking compartment. These embodiments intend to avoid overflowing events experienced when a user overfills the cooking chamber. The level of the cooking chamber contents is measured even before the heater is activated, and if there is a risk of overflowing, cooking is terminated before activation of the heater, avoiding likely overflowing events.
- [0008] In another embodiment of the invention, the control unit compares the limit cooking termination height saved in its memory according to the amount of coffee to be cooked and the value measured by the time of flight sensor, and terminates the cooking process when these values match.
- [0009] In another embodiment of the invention, the coffee machine comprises a shield glass protecting the time of flight sensor from dirt particles. The time

of flight sensor is affected from dirt particles such as coffee grounds, etc. on the shield glass, and the measured values are always below a given limit height independently of the levels in the cooking chamber. The control unit determines that the shield glass is dirty when the measured value is below the limit height. In a version of this embodiment, the control unit enables alerting the user upon determining the shield glass being dirty, and in another version, the control unit does not activate the heater in this case, preventing a defective cooking process.

[0010] In another embodiment of the invention, the time of flight sensor operates in wave length in the range of 900-1000 nm, particularly in the range of 930-950 nm, and more particularly of 940 nm. Light in the outer environment in said wave lengths is weakened by the humidity in air, therefore the time of flight sensor is thus enabled not to be affected from outer environmental factors.

[0011] The coffee machine of the invention enables both, cooking a desired number of cups of coffee, and avoiding losing taste and consistency in cooking coffees with different color tones.

[0012] A coffee machine realized to achieve the aim of the present invention is illustrated in the accompanying drawings, wherein:

[0013] Figure 1 is a perspective view of the coffee machine.

[0014] Figure 2 is a sectional view of the coffee machine in an embodiment of the invention.

[0015] The elements in the figures are numbered individually and the correspondence of these numbers are given hereinafter.

1. Coffee machine
2. Body
3. Cooking compartment
4. Cooking chamber
5. Heater
6. Control unit
7. Time of Flight Sensor
8. Cooking start button
9. Shield glass

- [0016] The coffee machine (1) comprises a body (2), a cooking compartment (3) provided in the body (2), a cooking chamber (4) in which the cooking process is performed, placed in the cooking compartment (3), a heater (5) provided on the base of the cooking compartment (3), and a control unit (6) starting the cooking process by activating the heater (5), terminating the cooking process by turning the heater (5) off, and enabling controlling the cooking process. In the coffee machine (1), the user fills Turkish coffee into the cooking chamber (4), places the cooking chamber (4) to the cooking compartment (3) and starts the cooking process. The control unit (6) starts the cooking process by activating the heater (5). Once the cooking process is completed, the control unit (6) turns the heater (5) off, terminating the cooking process.
- [0017] The coffee machine (1) of the invention comprises a time of flight sensor (7) provided in the body (2), enabling measuring the distance therefrom to the content of the cooking chamber (4) by being positioned so as to face the cooking compartment (3), and a control unit (6) enabling deciding to start and/or to terminate the cooking process according to the measured level value by the time of flight sensor (7). The time of flight sensor (7) enables measuring the distance between itself and the items disposed in the cooking compartment (3). The time of flight sensor (7) has a transmitter transmitting light and a receiver detecting the retroreflection of the light sent by the transmitter, and enables obtaining distance information by calculating the duration in which light travels between the transmitter and the receiver. When the cooking chamber (4) is placed in the cooking compartment (3), the distance is measured between the time of flight sensor (7) and the content of the cooking chamber (4), or if it is empty, the base of the cooking chamber (4). When the cooking process starts and the coffee water mixture in the cooking chamber (4) begins to foam and rise, the measured distance decreases.
- [0018] In an embodiment of the invention, the coffee machine (1) comprises a cooking start button (8) provided on the body (2), and the time of flight sensor (7) begins reading measurements once a user presses the cooking start button (8). Once the user places the cooking chamber (4) in the

cooking compartment (3) and presses the cooking start button (8), the time of flight sensor (7) begins reading measurements, measuring the amount of coffee, and of water if any, in the cooking chamber (4). If the measured value is above a predetermined limit overflow height, the control unit (6) does not start the cooking process. Thus, a cooking process with the risk of an overflow event upon activation of the heater (5), is not started at all and the user is alerted.

[0019] In another embodiment of the invention, the time of flight sensor (7) begins reading measurements upon placement of the cooking chamber (4) in the cooking compartment (3). Similar to the embodiment explained above, the time of flight sensor (7) begins reading measurements upon placement of the cooking chamber (4) in the cooking compartment (3), and likely overflow events are eliminated from the very beginning. Detection of placement of the cooking chamber (4) in the cooking compartment (3) can be performed by methods known in the art.

[0020] In another embodiment of the invention, the control unit (6) enables deciding to terminate the cooking process if depending on the amount of coffee to be cooked, the value measured by the time of flight sensor (7) is lower than a predetermined limit termination height. Different limit termination heights are saved in the memory of the control unit (6) according to the amount of coffee desired to be cooked and entered by using the coffee machine (1) interface by the user. During the cooking process, the time of flight sensor (7) continuously reads measurements, and the control unit (6) terminates the cooking process if the measured values are equal to or lower than the limit termination value corresponding to the amount of coffee desired to be cooked and entered to the control unit (6) by means of the interface.

[0021] In another embodiment of the invention, the coffee machine (1) comprises a shield glass (9) protecting the time of flight sensor (7) from the heat and steam released during cooking, and the control unit (6) decides that the shield glass (9) is dirty if the value measured by the time of flight sensor (7) is lower than a predetermined limit height. When the shield glass (9) is dirty, the measurements of the time flight sensor (7) are always below a

limit height due to light being blocked by the dirt particles on the shield glass (9). In this case, the control unit (6) can decide whether the shield glass (9) is dirty by considering the measured values. In a version of this embodiment, the control unit (6) enables alerting the user about dirtiness of the shield glass (9) once this is determined or upon completion of the current cooking process. The user is thus informed that he/she needs to clean the shield glass (9), and eliminating the necessity for the user to everlasting check the shield glass (9). In another version of this embodiment, the control unit (6) does not start the cooking process by not activating the heater (5) upon determining dirtiness of the shield glass (9). This enables preventing undesired circumstances likely to be caused by erroneous measurements such as overflowing events or cooking foul tasting coffee.

- [0022] In another embodiment of the invention, the time flight sensor (7) measures with light of a wave length in the range of 900 nm to 1000 nm, particularly measures with light of a wave length in the range of 930 nm to 950 nm, more particularly measures with light of a wave length of 940 nm. Water vapor in air absorbs light within said wave length range, therefore, in cases when the cooking compartment (3) is not flush with the cooking chamber (4) and the time of flight sensor (7) detects the outer environment, the measured value is enabled to be minimally affected from outer environment light by means of the selected wave lengths. This embodiment eliminates the necessity to position the cooking chamber (4) in the cooking compartment (3) in a light-proof manner, and enables a user to observe the cooking chamber (4) during the cooking process.
- [0023] In the coffee machine (1) of the invention, a desired amount of coffee is enabled to be cooked thanks to the time of flight sensor (7). Furthermore, it is not required to use an external sensor to determine whether the shield glass (9) is dirty.

## Claims

1. A coffee machine (1) **comprising** a body (2), a cooking compartment (3) provided in the body (2), a cooking chamber (4) in which the cooking process is performed, placed into the cooking compartment (3), a heater (5) provided on the base of the cooking compartment (3), and a control unit (6) starting the cooking process by activating the heater (5), terminating the cooking process by turning the heater (5) off, and enabling controlling of the cooking process, **characterized by** a time of flight sensor (7) provided in the body (2), enabling measuring the distance therefrom to the content of the cooking chamber (4) by being positioned so as to face the cooking compartment (3), **and by** the control unit (6) enabling deciding to start and/or to terminate the cooking process according to data received from the time of flight sensor (7).
2. A coffee machine (1) according to claim 1, **characterized by** a cooking start button (8) provided on the body (2), **and by** the time of flight sensor (7) beginning reading measurements once a user presses the cooking start button (8).
3. A coffee machine (1) according to claim 1, **characterized by** the time of flight sensor (7) beginning reading measurements upon placement of the cooking chamber (4) in the cooking compartment (3).
4. A coffee machine (1) according to any one of the preceding claims, **characterized by** the control unit (6), enabling deciding to terminate the cooking process if depending on the amount of coffee to be cooked, the value measured by the time of flight sensor (7) is lower than a predetermined limit termination height.
5. A coffee machine (1) according to any one of the claims 1 to 4, **characterized by** a shield glass (9) protecting the time of flight sensor (7) from the heat and steam released during cooking, **and by** the control unit (6) deciding dirtiness of the shield glass (9) if the value measured by the time of flight sensor (7) is lower than a predetermined limit height.
6. A coffee machine (1) according to claim 5, **characterized by** the control unit (6) enabling alerting the user if the shield glass (9) is determined to be dirty.
7. A coffee machine (1) according to claim 5 or 6, **characterized by** the control unit (6) not starting the cooking process by not activating the heater (5) if the

shield glass (9) is determined to be dirty.

8. A coffee machine (1) according to any one of the preceding claims,  
**characterized by** the time flight sensor (7) measuring with light of a wave length in the range of 900 nm to 1000 nm, particularly measuring with light of a wave length in the range of 9350 nm to 950 nm, more particularly measuring with light of a wave length of 940 nm.

Figure 1

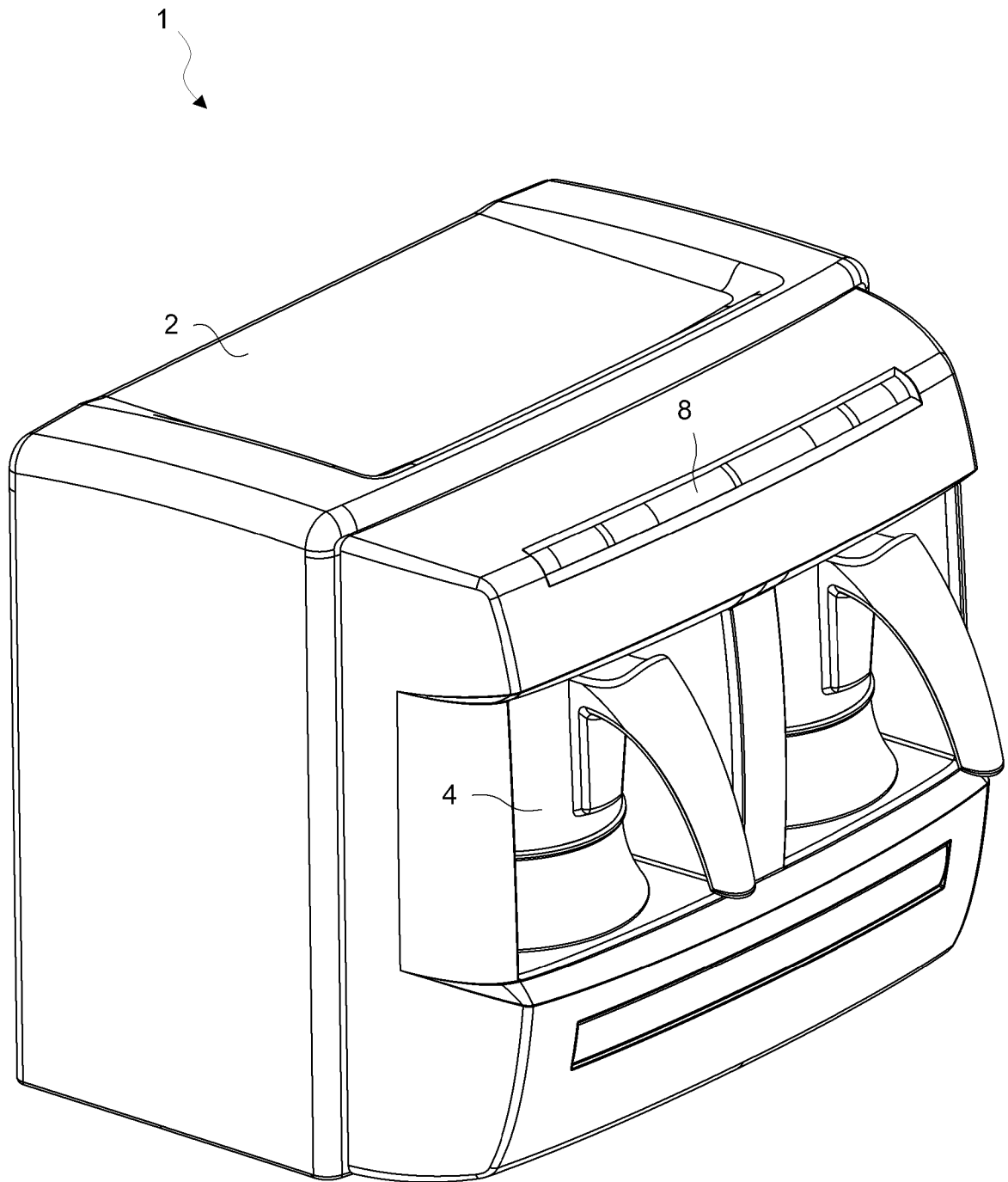
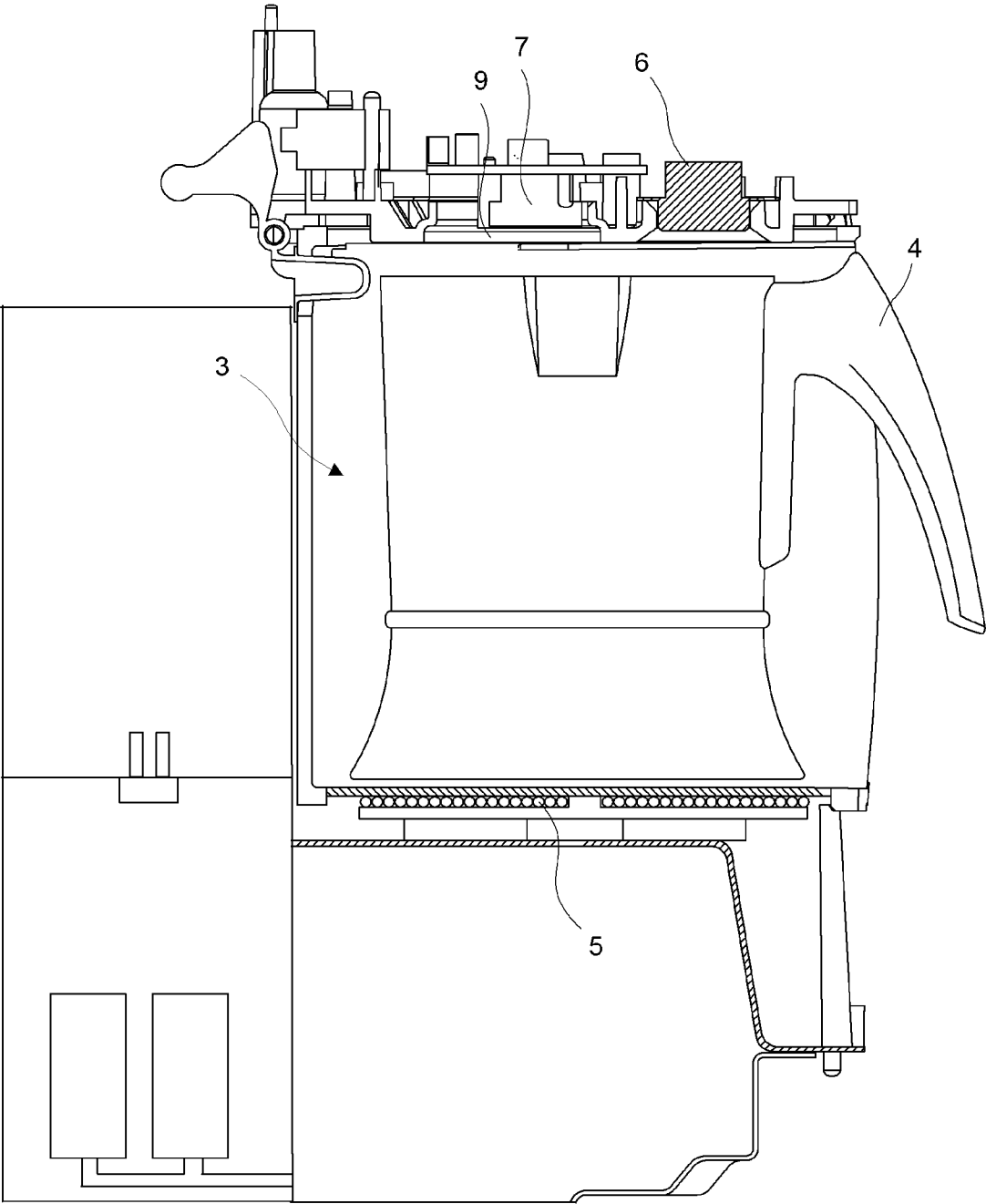


Figure 2



## INTERNATIONAL SEARCH REPORT

International application No  
PCT/EP2019/054639

A. CLASSIFICATION OF SUBJECT MATTER  
INV. A47J31/18 A47J31/00  
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

## B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)  
A47J

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data

## C. DOCUMENTS CONSIDERED TO BE RELEVANT

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Further documents are listed in the continuation of Box C.



See patent family annex.

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Date of the actual completion of the international search

2 April 2019

Date of mailing of the international search report

11/04/2019

Name and mailing address of the ISA/

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## INTERNATIONAL SEARCH REPORT

International application No  
PCT/EP2019/054639

| C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT |                                                                                                                                                                         |                       |
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| Category*                                            | Citation of document, with indication, where appropriate, of the relevant passages                                                                                      | Relevant to claim No. |
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# INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

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